Playing Unique Games on Certified Small-Set Expanders

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ABSTRACT

We give an algorithm for solving unique games (UG) instances whenever low-degree sum-of-squares proofs certify good bounds on the small-set-expansion of the underlying constraint graph via a hypercontractive inequality. Our algorithm is in fact more versatile, and succeeds even when the constraint graph is not a small-set expander as long as the structure of non-expanding small sets is (informally speaking) "characterized" by a low-degree sum-of-squares proof. Our results are obtained by rounding *low-entropy* solutions — measured via a new global potential function — to sum-of-squares (SoS) semidefinite programs. This technique adds to the (currently short) list of general tools for analyzing SoS relaxations for *worst-case* optimization problems.

As corollaries, we obtain the first polynomial-time algorithms for solving any UG instance where the constraint graph is either the *noisy hypercube*, the *short code* or the *Johnson* graph. The prior best algorithm for such instances was the eigenvalue enumeration algorithm of Arora, Barak, and Steurer (2010) which requires quasi-polynomial time for the noisy hypercube and nearly-exponential time for the short code and Johnson graphs. All of our results achieve an approximation of $1-\epsilon$ vs δ for UG instances, where $\epsilon > 0$ and $\delta > 0$ depend on the expansion parameters of the graph but are independent of the alphabet size.

CCS CONCEPTS

• **Theory of computation** → **Complexity theory and logic**; **Approximation algorithms analysis**.

KEYWORDS

Computational complexity theory, approximation algorithms, Sum of Squares algorithms, Unique games conjecture

ACM Reference Format:
Mitali Bafna, Boaz Barak, Pravesh K. Kothari, Tselil Schramm, and David Steurer. 2021. Playing Unique Games on Certified Small-Set Expanders. In *Proceedings of the 53rd Annual ACM SIGACT Symposium on Theory of Computing (STOC '21), June 21–25, 2021, Virtual, Italy.* ACM, New York, NY, USA, 14 pages. https://doi.org/10.1145/3406325.3451099

STOC '21, June 21–25, 2021, Virtual, Italy

ACM ISBN 978-1-4503-8053-9/21/06…$15.00
https://doi.org/10.1145/3406325.3451099

1 INTRODUCTION

The *Unique Games Conjecture (UGC)* [26] is a central open question in computational complexity and algorithms. In short, the UGC stipulates that distinguishing between almost satisfiable (value $\geq 1-\epsilon$) and highly unsatisfiable (value $\leq \epsilon$) instances of a certain 2-variable constraint satisfaction problem called *Unique Games* is NP-hard. The UGC is known to imply a vast number of hardness-of-approximation results in combinatorial optimization (e.g. Vertex Cover [31], Max Cut [27], constraint satisfaction problems [46], and Sparsest Cut [15]), but it is still not known whether the conjecture is true or false. In a significant recent breakthrough, Khot, Minzer, and Safra [30] (building on [11, 19, 29]) showed that it is NP-hard to distinguish $\frac{1}{2}$-satisfiable instances from ϵ-satisfiable instances. While [30]'s result leads to some hardness-of-approximation results [14], it is not sufficient to recover most of the striking consequences of the UGC. Moreover, regardless of the UGC's truth, there may be mild, natural conditions on instances that allow for polynomial-time algorithms for both the UG problem itself as well as "downstream" problems such as Max Cut.

The community-wide quest to potentially refute the UGC, as well as to understand conditions under which UG instances are easy, has produced numerous advances in the broader theory of algorithms over the past two decades. Notable examples include sophisticated graph partitioning tools [3, 37, 39, 40], local random walks and similar stochastic processes [2, 21], and perhaps most of all, new tools for analyzing and rounding semidefinite programs (SDPs). The groundbreaking result of Raghavendra [46] exposed a deep connection between the UGC and the performance of a semidefinite programming relaxation called the *basic SDP*, showing that the UGC implies that the basic SDP is the optimal polynomial-time algorithm for any constraint satisfaction problem. Efforts to refute the UGC thus naturally led to the study of more powerful SDPs such as the sum-of-squares (SoS) hierarchy [38, 45]. The study of SoS (and specifically SoS algorithms for unique games) has since blossomed in the algorithms community, leading to many algorithmic advances. These include the development of general techniques for analyzing and rounding SoS SDPs, such as global correlation rounding [12], and the proofs-to-algorithms perspective [7]. These techniques have in turn led to numerous algorithmic breakthroughs, for problems originating in high dimensional statistics (e.g. [6, 9, 16, 18, 23, 35, 36, 41]), quantum computation [10], statistical physics [24], and more [51].

In this work, we give new algorithms for a large family of structured instances of the Unique Games problem. Our algorithms are obtained via a novel analysis of Sum-of-Squares SDPs. Specifically, we define a new *global* potential function which is a proxy for the entropy of the distribution over non-integral SDP solutions to the UG instance, and we show that when the entropy of the SDP solution is low, it is easy to round. We are then able to control our potential function if the UG constraint graph satisfies certain properties. Our potential function offers an alternative to the *global correlation* function introduced by [12], which is one of very few known tools for analyzing SoS relaxations of worst-case problems. Using our new techniques, we show that polynomial-sized SoS relaxations solve Unique Games on graphs which were out of reach of previous techniques, including Short Code graphs, the Noisy Hypercube, and the Johnson graph.

To control our potential function, we exploit and deepen the connection between Unique Games and the related Small-Set Expansion problem. A graph is said to be a (δ, η)-small-set expander if all sets of measure at most δ have edge-expansion at least η, and the Small Set Expansion Hypothesis (SSEH) states that for each $\varepsilon > 0$ there exists a sufficiently small constant δ such that it is NP-hard to decide whether a given graph is a $(\delta, 1-\varepsilon)$-small set expander, or whether the graph contains a set of measure $\leq \delta$ with expansion $< \varepsilon$. A sequence of works in this area uncovered a fundamental relationship between the two problems [49, 50]. Our current state of knowledge regarding the relationship between these problems can be roughly summarized as follows. Raghavendra and Steurer [49] gave a reduction from the Small Set Expansion problem to Unique Games. Raghavendra, Steurer, and Tulsiani [50] reduced Unique games to Small Set Expansion, under the additional assumption that the UG constraint graph is itself a small-set expander. Though a reduction in the opposite direction (without this additional assumption) has been postulated, it is still not known whether UGC implies SSEH.

Both the reductions above were *worst-case reductions*, showing that if one problem is easy on *all* instances (or all instances of certain type, in the case of [50]'s work) then the other is also easy on all instances. In this work we show a "point-wise" reduction from UG instances on small-set expanders to the small-set expansion problem *within the sum-of-squares framework*. Specifically, we show that for a graph G for which SOS can certify small-set expansion, SOS can also solve *any* UG instance on G. More precisely, we show that our global SoS potential function always reflects the fact that the entropy is low in a small-set expander.

In addition to this pleasing qualitative statement, our result yields polynomial time algorithms for solving arbitrary Unique Games instances on algebraic families of constraint graphs such as the *noisy hypercube* [7, 32] and *short code* graphs [8] that have been extensively investigated in the context of constructing integrality gaps for UG and related problems. The quantitative guarantees of our rounding algorithm are substantially stronger than previously known and in particular we give the first polynomial-time algorithms for instances over these graphs in the UGC parameter regime of distinguishing between $1-\epsilon$ satisfiable and $\leq \delta$ satisfiable instances for small constant $\epsilon, \delta > 0$.

Our rounding technique is, in fact, more versatile and succeeds even when the constraint graph admits non-expanding sets so long as the structure of non-expanding small sets is (informally speaking) "understood" by the low-degree sum-of-squares proof system. Somewhat curiously, the main technical innovation in the recent proof of NP-hardness of the 2-to-1-Games problem due to Khot, Minzer and Safra [30] (building on [11, 19, 28, 29]) involves an exhaustive characterization of the structure of small non-expanding sets in algebraic families such as the *Johnson* and the *Grassmann* Graphs that establish the truth of the 2-to-1 conjecture. We show that their proof in fact yields a low-degree sum-of-squares certificate characterizing the non-expanding sets in the Johnson graph. Building on this, we obtain a polynomial time algorithm for solving arbitrary Unique Games instances when the constraint graph is the Johnson graph [1].

1.1 Our Results

We now formally state our results. Our first theorem shows that unique games is easy on graphs which are "certifiable small-set expanders." In order to state our theorem we first need to define certifiable small-set expanders. We use the well known relationship between *hypercontractivity* and small set expansion (e.g., [25]). This is a relation between a polynomial inequality derived from the graph and the combinatorial property that small sets have large expansion.

For a graph $G = (V, E)$ and $\lambda \geq 0$, we let $V_\lambda(G)$ denote the linear subspace of $\mathbb{R}^V$ that is spanned by the eigenvectors of G's normalized adjacency matrix that correspond to eigenvalues of value at least $1 - \lambda$. We say that G is (λ, C) *hypercontractive* if every $f \in V_\lambda(G)$ satisfies $\mathbb{E}_{v\sim V}[f_v^4] \leq C\,\mathbb{E}_{v\sim V}[f_v^2]^2$ [2]. It is known that if G is hypercontractive then subsets of size $\mathrm{poly}(\lambda)/C$ have expansion at least $\Omega(\lambda)$ and a certain converse was given in [7].

We say that G is (λ, C, D)-*certifiably hypercontractive* if G is (λ, C) hypercontractive and furthermore this fact is certifiable by a degree-D SoS proof (see Definition 6.1). Our main theorem shows that when a graph is certifiably hypercontractive, it is also a tractable constraint graph for unique games instances.[3]

Theorem 1.1 (Unique games on certifiable small-set expanders). *For every $C > 0$, $\lambda \in (0, 1)$, $D \in \mathbb{N}$ there exists a polynomial-time algorithm A such that if:*

- *G is (λ, C, D)-certifiably hypercontractive,* and
- *I is an affine unique games instance with constraint graph G, and $\mathrm{val}(I) = 1 - \varepsilon$, for $\varepsilon \leq \lambda^2/100$.*

Then $A(I)$ outputs an assignment to I with value at least $\frac{\varepsilon\lambda^4}{64C}$.

The algorithm is obtained by rounding the standard degree-D' SoS relaxation for unique games, where D' is a constant depending on $C, \varepsilon, \lambda, D$. The degree-$D'$ SoS relaxation for a unique games over

[1] The second largest eigenvalue of the Grassman graph's random walk matrix is already 1/2, and hence it is not an interesting constraint graph for the UGC regime of nearly satisfiable instances which we study in this work.

[2] This is also called *2 to 4 hypercontractivity*; we drop the "2 to 4" modifier as it is the only notion of hypercontractivity we use.

[3] To reduce clutter, we state many of our results with explicit numerical constants. We have made no attempt to optimize these.

constraint graph $G = (V, E)$ and alphabet Σ can be computed in $(|V| \cdot |\Sigma|)^{O(D')}$ time (see [52]).

Remark 1.2 (Completeness Gap vs Set Size)**.** The completeness bound in Theorem 1.1 is *independent* of the the parameter C corresponding to the set size. This is important, since, just as it works for expander graphs, the basic SDP can solve unique games instances on small set expanders if the completeness parameter can depend on the size of the sets that expand (see [4] and Theorem 1.1 of [48]). In contrast, obtaining a guarantee where the set-size δ is independent of the completeness such as the one in Theorem 1.1 inherently requires using higher levels of the SOS hierarchy, since there are known integrality gap instances for the basic SDP where the constraint graphs are certifiably hypercontractive (e.g., the short-code graph, see [8, Cor. 7.2] and [7]). As we discuss below, our algorithm solves such instances in polynomial time.

We prove Theorem 1.1 and give more precise quantitative bounds in Section 6. From this theorem, we are able to obtain corollaries for the Noisy Hypercube and the Noisy Short Code graphs, since the latter are known to have sum-of-squares certificates of small-set expansion via hypercontractivity [7].

Corollary 1.3 (Unique Games on the Noisy Hypercube). *For every $0.001 > \varepsilon > 0$ and $\frac{1}{4} > \alpha > 0$ there is a polynomial time algorithm A and a constant $\tau = \tau(\alpha, \varepsilon) > 0$, such that if I is an affine unique games instance over the α-noisy hypercube with $\mathrm{val}(I) \geq 1-\varepsilon$ then $A(I)$ outputs an assignment to I with value at least τ.*

Corollary 1.4 (Unique Games on the Noisy Short Code Graph). *There exists constant $\varepsilon_0 > 0$ such that for every $\varepsilon \in [0, \varepsilon_0)$, $\alpha \in (0, 1)$ there exists a polynomial-time algorithm A and a constant $\tau = \tau(\alpha, \varepsilon) > 0$, such that if I is an affine unique games instance over the α-noisy shortcode graph with $\mathrm{val}(I) \geq 1 - \varepsilon$, then $A(I)$ outputs an assignment to I with value at least τ.*

The value τ in both corollaries is of the form $\mathrm{poly}(\varepsilon) \exp(-c\sqrt{\varepsilon}/\alpha)$ for $c > 0$ a fixed constant. Crucially, τ is independent of the alphabet size of I. Though there was prior work giving SOS certificates for *specific* instances of this type (see Section 1.2), this is the first algorithm for *all* affine unique games instances over these graphs in the UGC parameter regime. We derive these corollaries (with more precise asymptotics) and give formal definitions of the relevant graphs in Section 7.

Finally, by extending our methods we are also able to obtain a result for the Johnson graph, (for n, ℓ, α, the (n, ℓ, α) Johnson graph has the vertices $\binom{[n]}{\ell}$ with $S \sim S'$ if $|S \cap S'| \geq (1 - \alpha)\ell$, see Definition 4.3) despite the fact that it is not a small-set expander.

Theorem 1.5 (Unique Games on the Johnson Graph). *For every $0.001 > \varepsilon > 0$, $\frac{1}{2} > \alpha > 0$, and integer $\ell \in \mathbb{N}$ with $\ell\alpha \in \mathbb{N}$, there is a polynomial-time algorithm A and a constant $\tau = \tau(\varepsilon, \alpha, \ell) > 0$ with the following guarantee: for $n \in \mathbb{N}$ sufficiently large, if I is an affine unique games instance over the (n, ℓ, α)-Johnson graph with $\mathrm{val}(I) \geq 1 - \varepsilon$, then $A(I)$ returns an assignment to I of value at least τ.*

The parameter τ is of the form $\mathrm{poly}\left(\frac{\varepsilon}{\binom{\ell}{r}\exp(c'r)}\right)$ for $r = c\varepsilon/\alpha$ and $c, c' > 0$ fixed constants; the runtime is polynomial in n with exponent that depends on ℓ, α, and ε. We prove Theorem 1.5 in Section 8, where we also give more precise quantitative guarantees.

Theorem 1.5 suggests that it may be possible to generalize Theorem 1.1 to establish that unique games is easy not only on graphs G that are certifiably small-set expanders, but even on graphs that are not small-set expanders but whose expansion profile has some "nice characterization" captured by low degree SoS proofs. Finding a formal notion of such a "nice characterization" is an interesting open question that can lead to a general understanding of the easy instances of unique games. It is also open whether the standard (i.e., non noisy) Boolean cube possesses such a characterization, and indeed it is not known whether constant-degree SoS (or any other polynomial-time algorithm) can solve unique games on the Boolean cube (see [1]).

1.2 Comparison with Prior Work

There has been an extensive prior literature on rounding sum-of-squares programs, solving unique games on restricted instances, and relating the unique-games and small-set expansion problems. In this section we discuss this literature and how our results relate to it.

1.2.1 Worst-Case Rounding Techniques for Higher-Degree SoS. Our main technical contribution is a new rounding technique that gives a new way to use higher-degree Sum-of-Squares relaxations for worst-case optimization problems. Despite the proliferation of uses of the sum-of-squares method in *average-case algorithm design* (e.g. [9, 41], see recent survey [20]), there are relatively few general techniques that harness the power of the higher-degree SoS relaxations for *worst-case* combinatorial optimization problems. The main examples for such techniques are [12]'s *global correlation rounding* and the generalization via *reweightings* developed in [10]. In this work we suggest a new way to round solutions to SoS relaxations by considering a potential function that measures the *entropy* of the distribution over non-integral SoS solutions via a proxy for the weighted collision probability of each variable's assignment. When the entropy is low (or collision probability is high), we show that it is easy to round to a solution with high value. We expect that this technique may find applications for other worst-case combinatorial optimization problems, including other CSPs, coloring, and the like.

1.2.2 Solving UG on Restricted Families of Constraint Graphs. Our work naturally fits into the long-standing investigation of efficient algorithms for various restricted families of instances of the Unique Games problem including expander graphs [5, 42], perturbed random graphs [34], and graphs with small "threshold rank" [3, 12, 22, 33]. In addition to yielding new algorithmic techniques and pointing out differences between hard CSPs such as 3-SAT (for which we do not know of any faster algorithm for such restricted families), such works constitute the best known "evidence" against the truth of the UGC. Our guarantees unify and extend these results by noting that each of these restricted families admit (special kinds of) low-degree sum-of-squares certificates of the constraint graph being a small-set expander. For low threshold-rank graphs, we get an algorithm as a direct corollary only when they are small-set expanders.

1.2.3 UG Algorithms for General Instances. The best currently-known algorithmic result for general instances of Unique Games is due to [3] and runs in time $\exp(n^{\mathrm{poly}(\varepsilon)})$ for all $1-\varepsilon$ satisfiable instances. This algorithm was shown to be captured by the SoS hierarchy (and extended to apply to other related problems) by [12, 22] . The algorithms of [3, 12, 22] have better running times when the constraint graph's adjacency matrix has few large eigenvalues: if there are at most r eigenvalues larger than $1-\mathrm{poly}(\varepsilon)$, then they run in time $\exp(r)$.

Our work improves upon the guarantees of [3, 12, 22] for instances which have super-logarithmically many large eigenvalues, yet have a constant degree sum-of-squares certificates of hypercontractivity. In particular, prior to our work, no polynomial-time algorithms were known for unique games instances over the noisy hypercube, noisy short code, and the Johnson graphs - the best known algorithm for the noisy-hypercube ran in quasi-polynomial time and for the noisy-short code ran in subexponential time.

1.2.4 Noisy Hypercube, Short Code, and Johnson Graphs. Starting with Khot and Vishnoi [32], the noisy hypercube (and more recently, the shortcode graph) has been intensely studied to construct integrality gaps for natural SDPs for UG. These works constructed specific instances of unique games over the noisy cube [48] and shortcode graph [8] that on one hand are very far from satisfiable but on the other hand cannot be certified to be so by certain weak SDP and LP hierarchies. [7] showed that the particular instances of [8, 32, 48] are in fact "easy" for SoS in the sense that they can be certified to be far from satisfiable by $O(1)$-degree SoS (see also [44]). However, the analysis of [7, 44] was tailored to the particular instances (specified by both the constraint graph *and* the edge constraints) of [8, 32, 48], and did not yield an algorithm for general instances over these constraint graphs. Specifically, [7] ported the analysis of the unsatisfiability proof from the works on integrality gaps into the SoS framework. However, this analysis was specific to the constructed instances. Moreover, [7] did not provide any *rounding algorithm* and is not directly applicable to analyzing satisfiable instances.

Our analysis of the SoS algorithm for the Johnson graph (Theorem 1.5) uses structural properties of the Johnson graph closely related to those shown by [28]. Similar structural properties of the *Grassman graph* have been exploited in the recent works [19, 30] to prove the so called "2-to-2 conjecture". This has been a recurring motif in works on unique games. In the noisy hypercube, short code, and now in the Johnson graph, structure that was exploited to prove soundness for reductions was later found useful in giving efficient algorithms for the same instances.

1.2.5 Reductions from UGC to SSEH. Raghavendra, Steurer and Tulsiani [50] (building on [49]) reduced the task of solving unique-games on small-set expanders to the small-set expansion problem (see also [53, Chap. 6]). Theorem 1.1 can be viewed as a "point-wise" version of their reduction for integrality gaps. Specifically, [50] gave a reduction which maps any unique-game instance (Π, G) (where G is a small set expander), into an instance G' of the small-set expansion problem, where G' is polynomially larger than G. In contrast, Theorem 1.1 shows that for *every* graph G, if $O(1)$-degree SoS certifies the small-set expansion of G then $O(1)$-degree SoS can also approximate unique games instances over *the same graph* G, which implies that if G a small-set expander and an integrality gap instance for the Degree D SoS relaxation of UG, then G is also an integrality gap instance for the Degree $\Omega(D)$ SoS relaxation for SSE. Our algorithm for the Johnson graph suggests that there might be a way to extend this result to a reduction from UG to SSE even when the constraint graph is *not* a small set expander but whose expansion profile has a nice characterization captured by SoS proofs, and hence is an easy instance of the small set expansion problem.

2 ORGANIZATION

In Section 4, we give a high-level overview of our algorithm and our proofs. In Section 5, we prove that if a certain potential function in the sum-of-squares relaxation has large value, then a simple algorithm produces assignments of value $\Omega(1)$. In Section 6 we prove that this potential is always large for certifiable small-set expanders, and in Section 7 we derive corollaries for the hypercube and short code graphs. Finally, in Section 8 we give the proof of Theorem 1.5 for the Johnson graph.

3 PRELIMINARIES AND NOTATION

For a (weighted) graph $G=(V,E)$, we use $(u,v)\sim E$ to denote an edge (u,v) sampled with probability proportional to its weight. We use A_G to denote the transition matrix of the random walk on G, $L_G = I - A_G$ to denote the Laplacian and π_G to denote the corresponding stationary distribution over V (we take π_G to be the distribution where each vertex is sampled proportional to the sum of weights on its incident edges (A_G might not have a unique stationary measure, for instance when G is bipartite or disconnected, but π_G is always a stationary measure of A_G.); we will drop the subscript when G is clear from context. It is easy to see that picking a random edge from E, is equivalent to picking a random vertex $v\sim\pi$ and a random neighbor w of v with probability proportional to the weight of the edge (w,v). For $v\in V(G)$, we use $\deg_G(v)$ to denote v's (weighted) degree inside G. If A is some probabilistic event or condition, we use $\mathbb{I}(A)$ to denote the indicator random variable of A (i.e., $\mathbb{I}(A)=1$ if A occurs and $\mathbb{I}(A)=0$ otherwise).

Definition 3.1 (Unique games)**.** A *unique games instance* is a pair $I=(G,\Pi)$ where $G=(V,E)$ is a graph and Π is a collection $\{\pi_{u,v}\}_{(u,v)\in E}$ such that $\pi_{u,v}$ is a permutation over some finite set Σ. The graph G is known as the *constraint graph* of I.

Given an instance $I=(G,\Pi)$ of unique games and an assignment $x\in\Sigma^V$ of values to the vertices of $G=(V,E)$, the *value* of x with respect to I is $\mathrm{val}_I(x) = \mathbb{E}_{(u,v)\sim E}\,\mathbb{I}(\pi_{u,v}(x_u)=x_v)$. The value of I is the maximum of $\mathrm{val}_I(x)$ over all $x\in\Sigma^V$. We may drop the subscript I when the instance is clear from context.

We say that (G,Π) is an *affine* unique games instance if Σ is an additive group and all the functions $\pi_{u,v}$ are of the form $\pi_{u,v}(x) = x - a_{u,v}$ for some $a_{u,v}\in\Sigma$. That is, all constraints correspond to $x_u - x_v = a_{u,v}$.

It is known that the UGC is equivalent to its restriction on affine instances [27]. In this paper we restrict attention to affine instances only. For the sake of simplicity, we will drop the qualifier "affine" in future discussion, but all of our results are for this family of constraints.

We will use the following standard result from approximation theory for approximating the step function using low degree polynomials:

Theorem 3.2 (Corollary of Theorem 4.5 in [17]). *Define $s_\alpha(x)$ to be the step function at $\alpha \in (0,1)$, so that $s_\alpha(x) = 0$ if $x < \alpha$ and 1 otherwise. Then for each $0 < \delta < \alpha$ and $\epsilon > 0$ there is a univariate polynomial of $p_\alpha^{\epsilon,\delta}$ of degree $O(\frac{1}{\delta}\log^2\frac{1}{\epsilon})$ such that*

(1) $|p_\alpha^{\epsilon,\delta}(x) - s_\alpha(x)| \le \epsilon$ for all $x \in [0, \alpha-\delta] \cup [\alpha+\delta, 1]$
(2) $0 \le p_\alpha^{\epsilon,\delta}(x) \le 1$ for all $x \in [0,1]$
(3) $p_\alpha^{\epsilon,\delta}$ is monotonically increasing on $(\alpha-\delta, \alpha+\delta)$.

Further, given axioms $A = \{x \ge 0\} \cup \{x \le 1\}$, there is an SoS proof that

$$A \vdash_{O(\frac{1}{\delta}\log^2\frac{1}{\epsilon})} \{0 \le p_\alpha^{\epsilon,\delta}(x) \le 1\}.$$

3.0.1 Sum of Squares Proofs. Given a set of axioms $\mathcal{A} = \{q_i = 0\}_i \cup \{g_j \ge 0\}_j$ for polynomials $q_i, g_j \in \mathbb{R}[x]$, we say that "there is a degree-d sum-of-squares proof that $f \ge h$ modulo $\mathcal{A}$" if $f = h + s + \sum_i c_i \cdot q_i + \sum_j r_j \cdot g_j$ with real polynomials $s, \{c_i\}_i, \{r_j\}_j \in \mathbb{R}[x]$ such that s and $\{r_j\}_j$ are sums of squares, and if the maximum degree among $s, \{c_i q_i\}_i, \{r_j g_j\}_j$ is at most d. We will use the notation $\mathcal{A} \vdash_d f(x) \ge h(x)$ to denote the existence of such an equality. We also sometimes use $f(x) \succeq h(x)$ to denote that the inequality is a SoS inequality.

3.0.2 Other Notation. We use the standard big-O and big-Ω notation. We will also use $f = \tilde{O}(x)$ to denote that there exists some c, C independent of x such that $\lim_{x\to\infty} \frac{f}{Cx\log^c x} \le 1$. For a positive integer k, we denote $[k] = \{1, \dots, k\}$ and $\binom{S}{\ell}$ to denote the set of unordered simple ℓ-element subsets of S. For a vector of variables x, we let $x^{\le D}$ denote the set of monomials of degree at most D in the variables. For a measure π on S and $f, g : S \to \mathbb{R}$, we use $\langle f, g\rangle_\pi = \mathbb{E}_{v\sim\pi} f(v)g(v)$ and the corresponding p-norms $\|f\|_{\pi,p} = (\mathbb{E}_{v\sim\pi} |f(v)|^p)^{1/p}$. For a function $f(x)$ and $k \in \mathbb{R}$, we will use $f^{\circ k}(x) = f(x)^k$ to denote the element-wise k-th power of f.

4 OVERVIEW OF OUR TECHNIQUES

We now describe our algorithm and give an overview of its analysis. Our algorithm is based on the SoS semidefinite programming (SDP) relaxation, and in particular its view as optimizing over *pseudo expectation* operators, see the surveys [13, 20, 47].

Given a unique games instance $I = (G, \Pi)$ over alphabet Σ, with $G = (V, E)$, the value of I can be computed by the following integer program over zero-one variables $\{X_{u,a}\}_{u\in V, a\in\Sigma}$:

$$\begin{aligned} \max_X \quad & \mathop{\mathbb{E}}_{(u,v)\in E} \sum_{a\in\Sigma} X_{u,a} X_{v,\pi_{uv}(a)} \qquad (1)\\ s.t. \quad & X_{u,a}^2 = X_{u,a} \quad \forall u \in V, a \in \Sigma \\ & X_{u,a}X_{u,b} = 0 \quad \forall u \in V, a \ne b \in \Sigma \\ & \sum_a X_{u,a} = 1 \quad \forall u \in V \end{aligned}$$

The variables $X_{i,a}$ are the 0/1 indicator variables that vertex $i \in V$ takes label $a \in \Sigma$. The objective function asks us to maximize the fraction of edge constraints satisfied. Our algorithm is obtained by considering the degree $D = O(1)$ SoS relaxation of the above program, obtaining a pseudo-expectation operator $\widetilde{\mathbb{E}} : X^{\le D} \to \mathbb{R}$, where $X^{\le D}$ is the set of all monomials in the X variables up to degree D, and $\widetilde{\mathbb{E}}$ satisfies the above equality constraints and the Booleanity constraints $\{X_{u,a}^2 = X_{u,a}\}$ as axioms. For brevity, we will refer to this set of axioms as $\mathcal{A}_I$, dropping the subscript when I is clear from context. The *value* of such a pseudo-expectation operator whose corresponding pseudodistribution is μ, with respect to the instance I is denoted by $\mathrm{val}_\mu(I) = \widetilde{\mathbb{E}}[\mathrm{val}_I(X)] = \widetilde{\mathbb{E}}[\mathbb{E}_{(u,v)\in E} \sum_{a\in\Sigma} X_{u,a}X_{v,\pi_{uv}(a)}]$. (Note that this is the pseudo expectation of a degree two polynomial in the variables $\{X_{u,a}\}$.)

4.1 Our Rounding Algorithm

The SoS SDP relaxation is standard, and the novelty of our work is in the *rounding* algorithm for it. A $(1-\varepsilon, \delta)$ rounding algorithm for the SoS relaxation is an algorithm that takes as an input an instance $I = (G, \Pi)$ and a pseudo-expectation operator $\widetilde{\mathbb{E}}$ (satisfying $\mathcal{A}_I$) of value at least $1-\varepsilon$ and outputs an assignment $x \in \Sigma^V$ with $\mathrm{val}_I(x) \ge \delta$. In this paper (and in the context of the UGC in general) we are interested in finding $(1-\varepsilon, \delta)$ rounding algorithms for ε, δ that are bounded away from zero by some constant which is independent of the alphabet size $|\Sigma|$.

Our rounding algorithm can be described as follows. We will define some low-degree polynomial $\Phi_\varepsilon^I : \mathbb{R}^{V\times\Sigma} \to [0, \infty)$ (which we call the "approximate shift partition potential" for reasons explained below). We then show (roughly speaking) the following three statements:

(1) There is a rounding algorithm that given an instance I and a pseudo-expectation operator $\widetilde{\mathbb{E}}$ such that $\widetilde{\mathbb{E}}[\mathrm{val}_I(X)] \ge 1-\varepsilon$ and $\widetilde{\mathbb{E}}[\Phi_\varepsilon^I(X)] \ge \delta$, outputs an assignment x for I with $\mathrm{val}_I(x) \ge \mathrm{poly}(\varepsilon, \delta)$.
(2) For every $I = (G, \Pi)$, if G is a $(\delta, 100\varepsilon)$-small-set expander, and if X is a random variable sampled from an actual distribution over vectors in $\{0,1\}^{V\times\Sigma}$ with expected value $1-\varepsilon$ for the integer program (1), then $\mathbb{E}[\Phi_\varepsilon^I(X)] \ge \mathrm{poly}(\delta)$.
(3) There is an $O(1)$-degree SoS *proof* for Statement 2.

Using the standard "SoS paradigm," the three steps above suffice to obtain algorithms for graphs that are certifiably small set expanders. For such graphs we can combine the expansion certificate with the SoS proof of Statement 2 to show that any pseudodistribution over X obtained as a solution of the SoS program will have to satisfy $\widetilde{\mathbb{E}}\,\Phi_\varepsilon^I \ge \Omega(1)$ and hence use the algorithm from Statement 1 to obtain an actual solution with value bounded away from zero.

In the case of the Johnson graph, which is not a small set expander, we have to work harder. In this case we use the characterization of non expanding sets in the Johnson graph to show that if the value is sufficiently large then the potential Φ_ε^I must be large on some ($o(1)$-sized) *subgraph* of the Johnson graph (itself a Johnson graph with different parameters). We solve for a partial assignment on this subgraph and iterate, and we are able to show that this process can continue until we have obtained an assignment with value independent of the alphabet size.

4.2 Rounding for Certified Small Set Expanders

Since our algorithm for the Johnson graph is more complex, we will start by describing our algorithm for certified small set expanders. In this section we will focus on the case that the pseudo expectation operator corresponds to an *actual distribution* and the graph G is simply a small set expander (with or without a certificate). This case is sufficient to illustrate the main ideas behind our algorithm. The full analysis is presented in Sections 5 and 6.

Throughout this section we fix an instance $I = (G, \Pi)$ of unique games, with $G = (V, E)$. We let μ be a distribution over strings X in $\{0,1\}^{V\times\Sigma}$ satisfying the constraints $\mathcal{A}_I$. We will also identify X with assignments in Σ^V and so write X_u for the unique element $s \in \Sigma$ such that $X_{u,s} = 1$.

For every vertex $u \in V$ and symbol $s \in \Sigma$, we define the following random variable

$$Z_{u,s} = \sum_{a\in\Sigma} X_{u,a}X'_{u,a+s} = \mathbb{I}(X_u - X'_u = s)\ ,$$

where X and X' are two independent samples from the distribution.

We think of Z_s as a subset of V, with $Z_{u,s}$ as the indicator variable for the membership of vertex u in Z_s. The $Z_{u,s}$'s satisfy partition constraints, hence they induce a partition of the graph into components on which the solutions X, X' agree up to a shift, so that $Z_{u,s} = 1$ when $X_u - X'_u = s$. We refer to this partition as the "shift partition." If we were to assign labels to the vertices arbitrarily, then each part Z_s in the partition would have size roughly $\approx \frac{1}{|\Sigma|}$. On the other hand, if there is a part in the partition of fractional size $\Omega(1)$, this means the labels of two independent assignments are more correlated than one would expect, in that they agree up to shift on a non-trivial fraction of vertices. This inspires our potential function.

We start by considering the following simplified version of our potential function:

Definition 4.1. For any $\beta \in (0,1)$, define the *shift-partition potential* to be the quantity

$$\Phi_\beta(X, X') = \sum_{s\in\Sigma}\left(\mathbb{E}_u\left(Z_{u,s}\cdot \mathbb{I}(\mathrm{val}_u(X) \geq \beta)\right)\right)^2,$$

for $\mathrm{val}_u(X)$ the "local objective" at u,
$\mathrm{val}_u(X) = \mathbb{E}_{v\sim u}\sum_{a\in\Sigma} X_{u,a}X_{v,\pi_{uv}(a)}$ where $v \sim u$ denotes a neighbor of u sampled according to the edge weight of (u, v).

This potential measures the average square size of components in the shift partition, where the indicator ensures that we only include vertices which satisfy at least a β fraction of incident edges. A convenient parameter setting will be to take $\beta = \varepsilon$.

4.2.1 Rounding from High Shift Partition Potential. If μ is an actual distribution with respect to an instance I, and $\mathbb{E}_\mu \Phi_\varepsilon(X, X') \geq \Omega(1)$, then the following simple algorithm (see also Algorithm 5.1) will find in expectation an assignment y for I with $\mathrm{val}_I(y) \geq \Omega(1)$:

(1) Pick $u_0 \in V$ uniformly at random.
(2) Sample $y_1, \ldots, y_V \in \Sigma$ independently by letting $\Pr[y_u = a] = \mathbb{E}[X_{u,a}|X_{u_0,0} = 1]$. (That is, y is sampled from the product distributions whose marginals correspond to $X|X_{u_0} = 0$.)

The intuition behind the above is as follows: When $\mathbb{E}[\Phi_\varepsilon(X, X')] \geq \delta$, then for a "typical" pair of independent assignments drawn from μ, there will be a subset of vertices S of measure $\geq \Omega(\delta)$ on which X and X' agree up to a shift in Σ. This implies that a random pair of vertices, will satisfy that the collision probability of the random variable $(X_v - X_u)$, i.e. $\Pr[X_u - X_v = X'_u - X'_v]$, is at least δ. Since we have symmetry over the labels, the distribution of $X_v - X_u$ is the same as the distribution of $(X_v|X_u = 0)$, hence we get that the collision probability of $(X_v|X_u = 0)$ is high. Since we now choose u_0 at random and condition the distribution on $X_{u_0} = 0$, we have that the marginal distribution of a random vertex has high collision probability. If in addition the value on the vertices with high collision probability is close to 1, we would immediately get that independent rounding gives high value on these vertices; this is because if a vertex's value *and* collision probability are high, the vertex's neighbors must also have high collision probability on the corresponding satisfying labels. Following this logic, independent rounding will satisfy a $\Omega(\delta^2)$-fraction of the edges incident on these vertices, so that in total we satisfy at least a $\Omega(\delta^3)$-fraction of the edges of the graph. The term $\mathbb{I}(\mathrm{val}_u(X) \geq \varepsilon)$ in the function Φ_ε ensures that the high-collision-probability vertices also correspond to high value vertices (since those vertices that have low value do not even contribute to the potential), and this suffices for us to make the above intuition go through. This argument is made formal in Section 5 (see Algorithm 5.1 and Theorem 5.3).

4.2.2 Low Degree Polynomials. The function Φ_β above cannot be used for rounding pseudo-expectation operators, because it is not a low degree polynomial in the variables X. To tackle this issue, we introduce the *approximate shift-partition potential*, replacing the high-degree indicator $\mathbb{I}(\mathrm{val}_u(X) \geq \beta)$ with an approximating low-degree polynomial:

Definition 4.2. For any $\nu, \beta \in (0,1)$, define the *approximate shift-partition potential* to be the quantity

$$\Phi_{\beta,\nu}(X, X') = \sum_{s\in\Sigma}\left(\mathbb{E}_u\left(Z_{u,s}\cdot p_{\beta,\nu}(\mathrm{val}_u(X))\right)\right)^2,$$

for $p_{\beta,\nu}(x)$ the degree-$\tilde{O}(1/\nu)$ polynomial which SoS-certifiably ν-approximates the indicator $\mathbb{I}[x \geq \beta]$ for $x \in [0,1]$ described in Theorem 3.2.

The function Φ^I_ε will be set as $\Phi_{\beta,\nu}$ for a suitable parameter setting $\beta = \varepsilon$ and $\nu = \mathrm{poly}(\varepsilon)$.

4.2.3 Small Set Expansion and the Shift Partition Potential. The $Z_{u,s}$ variables define a partition of the graph. Edges which cross this partition cannot be satisfied in *both* X and X' variables, since in an affine UG instance the labels of a satisfied edge's endpoints agree up to a shift: if (u, v) is an edge with u in the s shift component (that is, $X_u = X'_u + s$), and v in the t shift component ($X_v = X'_v + t$), then $X_u - X_v \neq X'_u - X'_v$ unless $s = t$. Therefore, the shift partition corresponds to a partition induced by removing the (on average) $\leq 2\varepsilon$ fraction of edges that are unsatisfied in at least one of the two solutions, X or X'. This means that if G is a $(\delta, 100\varepsilon)$-small-set expander, then on average the partition induced by $Z_{u,s}$ has parts of $\Omega(\delta)$ size. Since in an assignment of value $1 - \varepsilon$ there are at most $O(\varepsilon)$ vertices with local objective $\leq \varepsilon$, removing such vertices by introducing the indicators $\mathbb{I}[\mathrm{val}_u(X) \geq \varepsilon]$ removes at most $O(\varepsilon)$

edges and therefore the above reasoning is unaffected: the parts remain of size $\Omega(\delta)$, so that $\mathbb{E}[\Phi_\varepsilon] = \Omega(\delta)$. We make this intuition formal in Section 6 (see Theorem 6.2).

4.3 Johnson Graphs

The Johnson Graph is *not* a small-set expander. However, we are able to use its spectral structure to obtain a nontrivial approximation ratio. We start by formally defining this graph:

Definition 4.3 (Johnson Graph)**.** For any $1 > \alpha > 0$ and $\ell, q \in \mathbb{N}$ with $\alpha\ell \in \mathbb{N}$ and $n > \ell$, we define the (n, ℓ, α)*-Johnson graph* $J_{n,\ell,\alpha}$ to be the graph whose vertex set is $\binom{[n]}{\ell}$ and where edges are between pairs of vertices $U, V \in \binom{[n]}{\ell}$ if and only if $|U \cap V| = (1-\alpha)\ell$. We refer to α as the noise parameter (analogous to the α-noisy hypercube).

The (n, ℓ, α)-Johnson graph contains other Johnson graphs as subgraphs: consider the subgraph induced by vertices which contain some $S \subset [n]$ with $|S| < \ell$. We call such subgraphs $|S|$*-restricted subcubes*. It is not hard to see that such an r-restricted subcube contains at least an $\eta := (1-\alpha)^r$ fraction of its incident edges—this is because neighbors (U, V) differ in each element with probability $\approx \alpha$, and so for a random neighbor V of U, the chance that none of the elements of S are changed is $\approx (1-\alpha)^{|S|}$. Notice that when $r < O(\frac{c}{\alpha})$ and $r \ll \ell$, the fraction of internal edges in an r-restricted subcubes is at least $\eta \geq 1 - O(c)$.

[28] showed that in the Johnson graph, every non-expanding set that has expansion ε is correlated with some r-restricted subcube, for $r = O(\varepsilon/\alpha)$, that has expansion $O(\varepsilon)$. We show a "distribution-version" of this theorem: for any distribution over non-expanding sets, there exists an r-restricted subcube that is correlated with these sets in expectation. Moreover, we give an SoS proof of this fact (Theorem 8.5), so that the same statement holds for pseudodistributions too.

We then use this structure theorem to show that given a high value pseudodistribution for a unique games instance I, there must exist at least one r-restricted subcube, so that the approximate shift partition potential restricted to that subcube is high.

Lemma (Large potential on a subcube: special case of Lemma 8.9). *If I is a unique games instance on the (n, ℓ, α)-Johnson graph and X is sampled from a distribution over solutions with $\mathbb{E}[\mathrm{val}_I(X)] \geq 1 - \varepsilon$, then there exists an $O(\frac{\varepsilon}{\alpha})$-restricted subcube C such that the expected shift potential of the subgraph induced by C is at least $\delta = \delta(\ell, \varepsilon, \alpha) > 0$. Furthermore, this is certifiable by a degree-$\tilde{O}(1/\delta)$ SoS proof.*

The Johnson graph only has $\binom{n}{r} \leq \binom{n}{\ell}$ r-restricted subcubes, and so in $n^r = \mathrm{poly}(n)$ time we can enumerate over the cubes to find one cube C with a large shift-partition potential (i.e., satisfying $\mathbb{E}[\Phi_\varepsilon^C] \geq \delta$). We can then find a δ-satisfying solution for the internal edges of C by using our rounding algorithm (Algorithm 5.1). Since the fractional mass of C, $\mu(C) := \binom{n}{\ell-r}/\binom{n}{\ell} \approx \frac{\ell^r}{n^r}$, we only satisfy a negligible fraction of edges this way. On the other hand, since C is just a $o(1)$-fraction of the graph, the unique games instance restricted to the rest of the graph $\overline{C}$ must have high value too. Since the value remains high even after removing C, we may iteratively repeat this process to find a sequence of r-restricted subcubes $C_1, \ldots, C_T$, while ensuring that each cube C_t does not intersect too much with the previous subcubes $C_1, \ldots, C_{t-1}$. At each iteration, we fix an assignment on C_t satisfy an $\Omega(\delta^2)$-fraction of C_t's internal edges, which in turn is an $\Omega(\delta^2\eta)$-fraction of all edges incident on C_t; the remaining $(1-\delta^2\eta)$ fraction of edges incident on the cube (including outgoing edges) may be unsatisfied. But since the ratio of satisfied to unsatisfied edges incident on C_t is at least $\delta^2\eta$, the objective value drop (on the unassigned part of the graph) in every step is proportional to the fraction of edges we satisfy in that step. We repeat the process until the value drops by ε, so we end up satisfying an $\Omega(\delta^2\eta\varepsilon)$-fraction of all the edges.

Modulo the proof of the "large potential on subcube" Lemma (which will be a corollary of Lemma 8.9), this is nearly the complete argument. The only detail that remains is to apply the above lemma iteratively (we cannot simply apply it on $J \setminus C$ since that graph is not a Johnson graph) and to ensure that the subcubes we find at each iteration do not overlap too much. To handle both these issues, as we iterate we take additional measures. The full proof is in Section 8; see Algorithm 8.1 and Theorem 8.2.

5 ROUNDING INSTANCES WITH LARGE SHIFT POTENTIAL

In this section, we will show that when the objective value is large and the approximate-shift-partition potential Φ has large pseudoexpectation, then the Condition & Round Algorithm (Algorithm 5.1) succeeds in returning a good assignment for the unique games instance.

Algorithm 5.1 (Condition & Round)**.**
Input: A degree-D (for $D \geq 2$) shift-symmetric pseudodistribution (any pseudodistribution can be efficiently transformed into a shift-symmetric one without losing value) μ for a UG instance $I = (G = (V, E), \Pi)$ over alphabet Σ.
Goal: Return an assignment $x \in \Sigma^V$ satisfying $\Omega(1)$ fraction of the constraints in expectation.

Sample a random solution Y:

(1) Sample a vertex $u \sim \pi$ and condition on $X_u = 0$ to obtain the new marginals $\widetilde{\mathbb{E}}_\mu[\cdot \mid X_u = 0]$.
(2) Sample a solution Y by choosing each collapsed variable's labels independently according to its marginals: $Y_v \sim \widetilde{\mathbb{E}}_\mu[X_v \mid X_u = 0]$.

Recall the approximate shift-mass potential $\Phi_{\beta,\nu}(X, X')$ from Definition 4.2. We define the potential of a pseudo distribution μ to be the expectation of $\Phi_{\beta,\nu}$ over μ:

Definition 5.2 (Approximate shift mass potential of a pseudodistribution)**.** For a pseudodistribution μ of degree at least $2\deg(\Phi_{\beta,\nu})+2$, define the approximate shift mass potential of μ to be the quantity

$$\Phi_{\beta,\nu}(\mu) = \widetilde{\mathbb{E}}_{\mu}[\Phi_{\beta,\nu}(X, X')].$$

We will prove the following theorem:

Theorem 5.3. *Let $I = (G, \Pi)$ be an affine instance of Unique Games over the alphabet Σ. Let μ be a degree-$O(\deg(\Phi_{\beta,\nu}))$ shift-symmetric pseudodistribution satisfying the axioms $\mathcal{A}_I$ specified by program (1). If $\Phi_{\beta,\nu}(\mu) \geq \delta$, then on input μ Algorithm 5.1 runs in* *time* poly$(|V(G)|)$ *and returns an assignment of expected value at least* $(\delta - \nu)(\beta - \nu)$ *for* I.

While Algorithm 5.1 is randomized, we can derandomize it and obtain a deterministic polynomial-time algorithm with the same guarantee on the approximation factor. To derandomize we can use standard techniques such as the method of conditional expectations [54]. We will refer to such an algorithm as derandomized Condition & Round.

Proof of Theorem 5.3. Throughout this proof, we let μ be a pseudo-distribution satisfying the conditions of the theorem, and all pseudo-expectations are taken with respect to μ. Our overall strategy will be as follows: we will define an alternate potential function $\Psi(\mu)$, relate its value to $\Phi(\mu)$, and then show that when $\Psi(\mu)$ is large a single step of conditioning and independent rounding gives a large expected objective value.

To define our alternate potential, let us introduce some concise notation. For an event $\mathcal{E}$ whose indicator $\mathbb{I}(\mathcal{E})$ has degree at most $\deg(\mu)$ define $\widetilde{\Pr}[\mathcal{E}] = \widetilde{\mathbb{E}}[\mathbb{I}(\mathcal{E})]$. Similarly, for conditional probabilities, for events $\mathcal{E}$ and $\mathcal{F}$ with $\deg(\mathbb{I}(\mathcal{E} \wedge \mathcal{F})) \leq \deg(\mu)$, let $\widetilde{\Pr}[\mathcal{E} \mid \mathcal{F}] := \frac{\widetilde{\Pr}[\mathcal{E} \wedge \mathcal{F}]}{\widetilde{\Pr}[\mathcal{F}]}$. For simplicity of notation, when $\widetilde{\Pr}[\mathcal{F}] = 0$, we define $\widetilde{\Pr}[\mathcal{E} \mid \mathcal{F}] := 0$.

Now we define the conditioned shift potential $\Psi(\mu)$:

Definition 5.4. The *conditioned shift potential* of a degree-$D \geq 4$ pseudodistribution μ is given by

$$\Psi(\mu) := \mathop{\mathbb{E}}_{u,v\sim\pi}\left[\sum_{s\in\Sigma} \widetilde{\Pr_{\mu}}[X_v - X_u = s]^2 \cdot \widetilde{\mathbb{E}}[\mathrm{val}_v(X) \mid X_v - X_u = s]\right],$$

where π is the stationary measure on G and $\mathrm{val}_v(X)$ is the "local objective" at the vertex v, $\mathrm{val}_v(X) = \mathbb{E}_{w\sim v}[\widetilde{\Pr}_{\mu}[X \text{ satisfies } (v,w)]]$ for $w \sim v$ a neighbor of v sampled proportional to the weight on (v,w).

Roughly, the conditioned shift potential measures the average collision probability of the random variable $(X_u - X_v)$, but it gives more preference to those pairs (u,v) that have high local objective value.

We show that when $\Phi(\mu)$ is large, then $\Psi(\mu)$ is also large:

Lemma 5.5. *If the approximate shift mass potential of* μ *is large, then the conditioned shift potential of* μ *must be large as well:*

$$\Phi_{\beta,\nu}(\mu) \leq \frac{\Psi(\mu)}{\beta - \nu} + \nu.$$

Next, we show that when the conditioned shift potential is large, a single step of conditioning and rounding returns a solution of high objective value:

Lemma 5.6. *Let* $I = (G,\Pi)$ *be an affine instance of Unique Games over the alphabet* Σ. *Let* μ *be a degree-4 shift-symmetric pseudodistribution for* I. *When* $\Psi(\mu) \geq \delta$, *then the Condition & Round algorithm (Algorithm 5.1) returns a solution of expected value at least* δ.

We prove both these lemmas in the full version. Given the two lemmas, the first statement of the theorem clearly follows. □

6 CERTIFIABLE SMALL-SET EXPANDERS

In this section, we give an algorithm for unique games on certifiable small set expander graphs, when the certificate is via 2-to-4 hypercontractivity. To state our theorem, we will require the following definition:

Definition 6.1. (Certifiable 2 to 4 hypercontractivity) For $C \in \mathbb{R}_+$, $\lambda \in (0,2)$, and $D \geq 2$ an integer, a graph $G = (V,E)$ is said to be (λ, C, D)-*certifiably 2 to 4 hypercontractive* if for any $f : V \longrightarrow \mathbb{R}$,

$$\vdash_D \quad \|\Pi_\lambda f\|_{\pi,4}^4 \leq C \cdot \|f\|_{\pi,2}^4,$$

where $\|f\|_{\pi,p} = (\mathbb{E}_{v\sim\pi} f(v)^p)^{1/p}$, and Π_λ is the projection to the right eigenspace of eigenvalues at most λ of G's normalized Laplacian.

We will also say that a graph is a (ε, δ, D)-certifiable SSE if there is a degree-D SoS proof that sets of size $\leq \delta$ have expansion at least ε.

Our main theorem is the following (more fleshed out version of Theorem 1.1):

Theorem 6.2. *For any* (λ, C, D)*-certifiable 2 to 4 hypercontractive graph* G *and for all* $\varepsilon < \frac{1}{100}\lambda^2$, *given a degree-*$(D + \tilde{O}(C/\varepsilon\lambda^4))$ *shift-symmetric pseudodistribution* μ *of value* $\geq (1-\varepsilon)$ *for an affine Unique Games instance* $I = (G,\Pi)$ *on* G, *Algorithm 5.1 runs in time* poly$(|V(G)|)$ *and outputs an assignment with expected value at least* $\frac{\varepsilon\lambda^4}{64C}$.

Proof. We start with the fact that a graph which is certifiably 2 to 4 hypercontractive is also a certifiable small-set expander. This was shown in [7], but we will state and use stronger guarantees about the form of the certificate which were implicit in their proof.

Lemma 6.3 (Lemma 6.7 in [7]). *If* $G = (V,E)$ *is* (λ, C, D)*-certifiably 2 to 4 hypercontractive,* G *is a* $(\lambda/2, \lambda^4/(16C), D)$*-certifiable small-set expander: for any* $f : V \longrightarrow \mathbb{R}$,

$$\left\{\|\Pi_\lambda f\|_{\pi,4}^4 \leq C \cdot \|f\|_{\pi,2}^4\right\} \cup \left\{f(v)^2 = f(v)\right\}_{v\in V} \cup \left\{\mathop{\mathbb{E}}_{\pi} f \leq \frac{\lambda^4}{16C}\right\}$$
$$\vdash_{4+D} \quad \langle f, Lf\rangle_\pi \geq \frac{\lambda}{2}\mathop{\mathbb{E}}_{\pi}[f],$$

Where Π_λ *is the projector to the right eigenspace of eigenvalue* $\leq \lambda$ *in* G*'s normalized Laplacian. Further,*

$$\left\{\|\Pi_\lambda f\|_{\pi,4}^4 \leq C\|f\|_{\pi,2}^4\right\} \cup \{0 \leq f(v) \leq 1\}_{v\in V}$$
$$\vdash_{4+D} \ \langle f, Lf\rangle_\pi \geq \frac{\lambda}{2}\mathop{\mathbb{E}}_{\pi}[f] + c\left(\frac{\lambda^4}{16C}\mathop{\mathbb{E}}_{\pi}[f] - \mathop{\mathbb{E}}_{\pi}[f]^2\right) + B(f)$$

For c *a positive constant and* $B(f) = 2(\mathbb{E}_\pi[f^{\circ 2} - f]) + \langle f^{\circ 3} - f, \Pi_\lambda f\rangle_\pi$.

Letting $\alpha := \frac{\lambda}{2}$ and $\gamma := \frac{\lambda^4}{16C}$, our assumptions together with Lemma 6.3 give us a small-set expansion certificate of the following form:

$$SSE_{\alpha,\gamma}(G) \equiv \{0 \leq f(u) \leq 1\}_{u\in V}$$
$$\vdash_{D+4} \ \langle f, Lf\rangle_\pi \geq \alpha\mathop{\mathbb{E}}_{\pi}[f] + c_1 \cdot \left(\gamma\mathop{\mathbb{E}}_{\pi}[f] - \mathop{\mathbb{E}}_{\pi}[f]^2\right) + B(f), \quad (2)$$

for c_1 a positive constant, $B(f) = 2(\mathbb{E}_\pi[f^{\circ 2} - f]) + \langle f^{\circ 3} - f, Pf\rangle_\pi$ and P a projection operator.

Next, we will show that if a graph has such a certificate of small-set expansion, then one can also obtain a lower bound on the approximate shift potential $\Phi_{\beta,\nu}(X,X')$ (whose definition we now recall), which gives a condition under which we can round. Theorem 3.2 guarantees the existence of a family $P_{\beta,\nu}$ of degree-$\tilde{O}(1/\nu)$ polynomials SoS-certifiably which approximate $\mathbb{I}[x \geq \beta]$ within an additive ν in the intervals $[0, \beta-\nu] \cup [\beta+\nu, 1]$. Fix $p \in P_{\beta,\nu}$ to be one such polynomial. The functions $\{f_s : V \to \mathbb{R}[X,X']\}_{s\in\Sigma}$ defined such that

$$f_s(u) = \mathbb{I}(X_u - X'_u = s) \cdot p(\mathrm{val}(X_u)) \tag{3}$$

give disjoint *approximate* vertex subsets of G (approximate only because p is not exactly an indicator). Recall the definition of the approximate shift-mass potential (Definition 4.2):

$$\Phi_{\beta,\nu}(X,X') = \sum_{s\in\Sigma}\left(\mathop{\mathbb{E}}_u f_s(u)\right)^2$$
$$= \sum_{s\in\Sigma}\left(\mathop{\mathbb{E}}_u \left(\mathbb{1}(X_u - X'_u = s)\cdot p(\mathrm{val}(X_u))\right)\right)^2.$$

Edges crossing this partition must be unsatisfied in either X or X' (see the discussion in Section 4). In a certifiable small-set expander with large objective value, this partition cannot cut too many edges, and therefore its pieces must be large. We will make this formal via the following lemma:

Lemma 6.4. *Let I be a unique games instance over a graph $G = (V,E)$ in which functions $f : V \to [0,1]$ with support $\leq \gamma$ are SoS-certifiably α-expanding via the following certificate:*

$$SSE_{\alpha,\gamma}(G) :\equiv \{0 \leq f(v) \leq 1\}_{v\in V}$$
$$\vdash_D \left\{\langle f, Lf\rangle_\pi \geq \alpha \mathop{\mathbb{E}}_\pi[f] + c\cdot\left(\gamma\mathop{\mathbb{E}}_\pi[f] - \mathop{\mathbb{E}}_\pi[f]^2\right) + B(f)\right\},$$

where $B(f) = 2(\mathbb{E}_\pi[f^{\circ 2} - f]) + \langle f^{\circ 3} - f, Pf\rangle_\pi$, c is a fixed positive constant and P is a projection operator.

Then we have that, for all $\beta \in (0,1)$, $\nu \in (0, \frac{1}{3}(1-\beta))$, and $\eta \in \mathbb{R}^+$, there is an SoS lower bound on the approximate shift mass potential $\Phi_{\beta,\nu}$:

$$\mathcal{A}_I \cup \{p \in P_{\beta,\nu}\} \cup \mathrm{SSE}_{\alpha,\gamma}(G)$$
$$\vdash_{D+\tilde{O}(1/\nu)} \Phi_{\beta,\nu}(X,X') \geq \gamma\left(1 - \frac{\mathrm{viol}(X)}{1-\beta-\nu} - \nu\right) + K^{\alpha,\eta}_{\beta,\nu}(X,X'),$$

where $\mathcal{A}_I$ are the axioms defined for I by program (1),
$\mathrm{viol}(X) = 1 - \mathrm{val}(X)$ *is the fraction of constraints X violates, and* $K^{\alpha,\eta}_{\beta,\nu}(X,X')$
$= c'\cdot\left(\alpha - (4+\alpha+\eta)\left(\frac{\mathrm{viol}(X)}{1-\beta-\nu}+\nu\right) - \frac{1}{2\eta} - (\mathrm{viol}(X)+\mathrm{viol}(X'))\right)$
for $c' \in \mathbb{R}^+$.

We give the proof in the full version. Informally, the quantity $K^{\alpha,\eta}_{\beta,\nu}(X,X')$ can be made non-negative when the fraction of violations $\mathrm{viol}(X)$ and $\mathrm{viol}(X')$ are small relative to the expansion α.

From equation (2) and Lemma 6.4, we may choose $\beta = \varepsilon \leq .01$, $\nu = \varepsilon\gamma$, and $\eta = \frac{1}{2\sqrt{\varepsilon}}$, and the conditions of our theorem imply that we have a degree-$(D + \tilde{O}(1/\varepsilon\gamma))$ sum-of-squares proof that

$$\Phi_{\varepsilon,\varepsilon\gamma}(X,X') \geq \gamma\left(1 - \frac{\mathrm{viol}(X)}{1-\varepsilon-\varepsilon\gamma} - \varepsilon\gamma\right) + K^{\alpha,\sqrt{1/4\varepsilon}}_{\varepsilon,\varepsilon\gamma}(X,X'). \tag{4}$$

In order to apply our rounding Theorem 5.3, we require that the pseudoexpectation $\tilde{\mathbb{E}}[\Phi_{\varepsilon,\varepsilon\gamma}(X,X')]$ is large, where $\tilde{\mathbb{E}}$ is the pseudoexpectation operator corresponding to the pseudodistribution μ given to us. Since by assumption $\tilde{\mathbb{E}}[\mathrm{viol}(X)] = \tilde{\mathbb{E}}[\mathrm{viol}(X')] \leq \varepsilon$, $\tilde{\mathbb{E}}$ has degree $(D + \tilde{O}(C/\varepsilon\lambda^4)) = D + \tilde{O}(1/\varepsilon\gamma)$ and $\tilde{\mathbb{E}}$ satisfies $\mathcal{A}_I$, we take the pseudoexpectation of (4) to get

$$\tilde{\mathbb{E}}\left[\Phi_{\varepsilon,\varepsilon\gamma}(X,X')\right] \geq \gamma\left(1 - \frac{\varepsilon}{1-\varepsilon-\varepsilon\gamma} - \varepsilon\gamma\right) + \tilde{\mathbb{E}}\left[K^{\alpha,\sqrt{1/4\varepsilon}}_{\varepsilon,\varepsilon\gamma}(X,X')\right]. \tag{5}$$

We show now that for our chosen parameters, $\tilde{\mathbb{E}}[K^{\alpha,1/\sqrt{4\varepsilon}}_{\varepsilon,\varepsilon\gamma}(X,X')] \geq 0$. Expanding the expression for K and using our bound on $\tilde{\mathbb{E}}[\mathrm{viol}(X) + \mathrm{viol}(X')]$,

$$\tilde{\mathbb{E}}\left[K^{\alpha,1/\sqrt{4\varepsilon}}_{\varepsilon,\varepsilon\gamma}(X,X')\right]$$
$$\geq c'\cdot\left(\alpha - \left(4+\alpha+\frac{1}{2\sqrt{\varepsilon}}\right)\left(\frac{\varepsilon}{1-\varepsilon-\varepsilon\gamma}+\varepsilon\gamma\right) - \sqrt{\varepsilon} - 2\varepsilon\right)$$
$$\geq c'(\alpha - 5\sqrt{\varepsilon}),$$

where to obtain the final inequality we have used that $\gamma < \frac{1}{2}$, $\varepsilon < \frac{1}{25}$, and $\alpha < 1$. Since $\alpha = \frac{\lambda}{2} \geq 5\sqrt{\varepsilon}$ by assumption and since $c' \in \mathbb{R}_+$, this quantity is non-negative.

Returning to (5) and simplifying with our upper bounds $\varepsilon < \frac{1}{25}$, $\gamma < \frac{1}{2}$, we have that

$$\tilde{\mathbb{E}}\left[\Phi_{\varepsilon,\varepsilon\gamma}\right] \geq \frac{3}{4}\gamma.$$

Applying Theorem 5.3, we conclude that conditioning and rounding a degree-$(D + \tilde{O}(C/\varepsilon\lambda^4))$ pseudodistribution according to Algorithm 5.1 results in a solution of expected value $\geq (\frac{3}{4}\gamma - \varepsilon\gamma)(\varepsilon - \gamma\varepsilon) \geq \frac{1}{4}\varepsilon\gamma = \frac{\varepsilon\lambda^4}{64C}$, as desired. □

7 UG ON NOISY-HYPERCUBE AND SHORT-CODE GRAPHS

Here, we derive two corollaries of Theorem 6.2: we show that polynomial-time sum-of-squares relaxations solve Unique Games on the noisy hypercube graph and the short-code graph. These results follow easily by combining our results with the prior results of Barak et al. [7], who showed that these graphs are certifiably 2 to 4 hypercontractive in sum-of-squares degree 4.

We first treat the noisy hypercube:

Definition 7.1 (Noisy Hypercube Graph)**.** For each $\varepsilon \in [0,1]$ and $d \in \mathbb{N}_+$, the ε-noisy d-dimensional hypercube is the graph on $\{\pm 1\}^d$, with weighted edges $\{w_{uv}\}_{u,v\in\{\pm1\}^d}$ where $w_{u,v} = (\varepsilon)^{(d-\langle u,v\rangle)/2}(1-\varepsilon)^{(d+\langle u,v\rangle)/2}$.

Motivated by breaking known Unique Games integrality gaps, the work of [7] showed that the classical proof of hypercontractivity for the noisy hypercube (see e.g. [43]) can be recast as a degree-4 sum-of-squares proof.

Theorem 7.2 (Noisy-Hypercube Certificate ([7], Lemma 5.1)). *Suppose G is the d-dimensional α-noisy hypercube. Then for any $t \in [d]$, G is $(1-(1-2\alpha)^t, 9^t, 4)$-certifiably 2 to 4 hypercontractive.*

In the same work, Barak et al. [7], building on [8], noted that the same argument shows that the short code graph is also SOS-certifiably 2 to 4 hypercontractive.

Definition 7.3 (Short Code Graph). For each $d < n \in \mathbb{N}_+$, the (d,n)-shortcode graph is a graph whose vertex set is the set of degree-d polynomials over $\mathbb{F}_2^n$ and with edges between each pair of polynomials p,q such that $p-q$ is a product of d linearly independent affine forms. For any $\alpha \in [0,1)$, the *α-noisy (d,n)-shortcode graph* is the graph with the random walk transition matrix $(G_{d,n})^{1+\alpha 2^d}$.

Remark 7.4. The noisy version of the short code is qualitatively similar to the noisy hypercube, since the transition probabilities in the α-noisy n-dimensional cube are similar to performing an αn-step random walk on the hypercube graph. In [8], a different notion of noise is used, where they instead consider the graph with adjacency matrix $\exp(-\alpha 2^d(I-G_{d,n}))$; our results can be reformulated for this notion of noise as well.

Theorem 7.5 (Short-Code Certificate [7]). *Suppose G is the α-noisy (d,n)-shortcode graph, and let $\ell = \lfloor \eta \cdot 2^d \rfloor$ for η a universal constant. Then for any $t \in [\ell]$, G is $(1-(1-t2^{-d})^{1+\alpha 2^d}, 9^t, 4)$-certifiably 2 to 4 hypercontractive.*

Combining Theorem 6.2 with these results, we show that Unique Games instances on the Noisy Hypercube and Short Code graphs are easy.

Theorem 7.6 (UG on Noisy-Hypercube, re-statement of Corollary 1.3). *For every $\varepsilon \in [0, \frac{1}{400})$, $\alpha \in (0, \frac{1}{4})$, and $d \in \mathbb{N}$ sufficiently large, there exists an algorithm A with the following guarantee: if $I = (G,\Pi)$ is an instance of Unique Games on the d-dimensional α-noisy hypercube G with $\mathrm{val}(I) \geq 1-\varepsilon$, then in time $|V(G)|^{\mathrm{poly}(\tau,1/\varepsilon)}$, $A(I)$ returns an $\Omega(\varepsilon^3/\tau)$-satisfying assignment for I for $\tau = \exp(O(\sqrt{\varepsilon}/\alpha))$.*

Theorem 7.7 (UG on Short-Code, re-statement of Corollary 1.4). *There exist $\varepsilon_0 \in \mathbb{R}_+$ such that for every $n \in \mathbb{N}$ sufficiently large and $d \in \mathbb{N}$ with $2d < n$, $\varepsilon \in (0,\varepsilon_0)$, and $\alpha \in (0,1)$, there is an algorithm A with the following guarantee: if (G,Π) is an instance of Unique Games on the alpha-noisy (d,n)-shortcode graph with $\mathrm{val}(G,\Pi) \geq 1-\varepsilon$, then in time $|V(G)|^{\mathrm{poly}(1/\varepsilon,\tau)}$, $A(G,\Pi)$ returns a solution of value $\Omega(\varepsilon^3/\tau)$ for (G,Π) for $\tau = \min\left(\exp(O(\sqrt{\varepsilon}/\alpha)), \exp(O(\sqrt{\varepsilon}2^d))\right)$.*

8 JOHNSON GRAPHS

In this section we'll prove that Algorithm 8.1 succeeds in producing an assignment with good value for unique games instances of sufficiently high value over the Johnson graph.

Algorithm 8.1 (Unique Games on the Johnson Graph). Takes as input an affine UG instance on a (n,ℓ,α)-Johnson graph $I = (J,\Pi)$ over labels Σ with $\mathrm{val}(I) = 1-\varepsilon$, returns a $\Omega_{\varepsilon,\alpha,\ell}(1)$ satisfying assignment.

(1) Fix $r = \left\lfloor \frac{32\varepsilon}{\alpha} \right\rfloor$, $\delta(\eta) := \frac{\eta}{\exp(cr)\binom{\ell}{r}}$ for all $\eta \in [0,1]$ and $D = \widetilde{O}(\frac{1}{\delta(\varepsilon)})$, for $c > 0$ a universal constant. Fix $\mathcal{A}_I$ to be the set of unique games axioms/integer program over the instance I (Program 1).
(2) Solve the degree-D SoS SDP relaxation for the integer program $\mathcal{A}_I$ and symmetrize the pseudodistribution over additive shifts (see Section 3 in full version) to get μ_0. Set $j=1$.
(3) While the SDP value $\mathrm{val}_{\mu_{j-1}}(I) \geq 1-2\varepsilon$:
   (a) For any $r' \leq r$, find an r'-restricted subcube C_j (induced subgraph of J, defined formally in Definition 8.3) with high Condition&Round value[4]: $\text{CR-val}_\mu(C_j) \geq \delta(\eta_{j-1})$ for $\eta_{j-1} = 1-\mathrm{val}_{\mu_{j-1}}(I)$.
   (b) Let S_j be a subgraph of C_j induced by the set of vertices that have not been previously assigned by any partial assignment $f_k, k<j$. Perform derandomized Condition&Round on $V(S_j)$ to get a partial assignment f_j[5].
   (c) Rerandomize the pseudodistribution μ_{j-1} on S_j to get μ_j: Make the marginal distribution over the assigned vertices uniform and independent of other vertices, that is, for all degree $\leq D$ monomials define $\widetilde{\mathbb{E}}_{\mu_j}$ as follows,
$$\widetilde{\mathbb{E}}_{\mu_j}[X_{h_1,a_1}\cdots X_{h_t,a_t}X_{u_1,b_1}\cdots X_{u_m,b_m}] := \frac{1}{|\Sigma|^t}\widetilde{\mathbb{E}}_{\mu_{j-1}}[X_{u_1,b_1}\cdots X_{u_m,b_m}],$$
   where $\{(h_1,a_1),\ldots,(h_t,a_t)\} \in (V(S_j)\times\Sigma)^t$ and $\{(u_1,b_1),\ldots,(u_m,b_m)\} \in ((\binom{[n]}{\ell}\setminus V(S_j))\times\Sigma)^m$.
   (d) Increment j.
(4) Output any assignment $f : V \to \Sigma$ that agrees with all partial assignments f_j considered above.

We show that this algorithm returns a solution with value independent of the alphabet size.

Theorem 8.2. *For every $\varepsilon \in [0, \frac{1}{2000})$, $\alpha \in \mathbb{Q}$ with $\alpha < \frac{1}{2}$, $\ell \in \mathbb{N}$ with $\alpha\ell \in \mathbb{N}$, and integers k,n sufficiently large, Algorithm 8.1 has the following guarantee: if I is an instance of affine Unique Games on the (n,ℓ,α)-Johnson graph J with alphabet size $|\Sigma| = k$ and $\mathrm{val}(I) = 1-\varepsilon$, then in time $|V(J)|^{\mathrm{poly}(\binom{\ell}{r},1/\varepsilon)}$, $A(I)$ returns an $\Omega\left(\frac{\varepsilon^3}{\exp(O(r))\binom{\ell}{r}^2}\right)$-satisfying assignment for I for $r = O(\varepsilon/\alpha)$.*

The proof of Theorem 8.2 requires some additional ideas beyond that of Theorem 6.2, as the Johnson graph is not a small-set expander. Nevertheless, we can characterize the structure of all the non-expanding sets, that is, we can prove that any non-expanding set must be large inside some canonical subgraphs. Using this characterization we prove that the above algorithm succeeds in finding a good assignment. The proof of our main theorem proceeds in the following steps:

(1) We first prove a structure theorem (Theorem 8.5) for non-expanding sets of the Johnson graph, similar to the theorem in [28]. We show an SoS proof of the fact that every non-expanding set must be large when restricted to subcubes of the Johnson graph (Definition 8.3).
(2) Using the structure theorem, in Lemma 8.7 we first lower bound the global shift-partition potential $\Phi_{\beta,\nu}(X,X')|_C$ as a function of the violations of X and X'. Roughly the global

[4]The quantity $\text{CR-val}_\mu(C)$ corresponds to the expected value obtained when Algorithm 5.1 is performed on the subgraph C and is formally defined in Definition 8.8.
[5]As noted earlier, derandomization produces an assignment that satisfies $\text{CR-val}_{\mu_{j-1}}(S_j)$-fraction of edges and can be performed in polynomial time using the method of conditional expectations.

shift-partition potential $\Phi_{\beta,\nu}(X,X')|_C$ corresponds to the shift-component squared sizes when restricted to the subcube C (see Definition 8.6). This lemma follows the same outline as that of Lemma 6.4 for certifiable small-set expanders.

(3) In the next step (Lemma 8.9), we show that given a pseudodistribution μ with objective value $1-\varepsilon$ for unique games over the Johnson graph, one can find a subcube C that has high global shift-partition potential. We then relate the global shift-partition potential to the shift-partition potential on the subgraph induced by C, $\widetilde{\mathbb{E}}_{\mu|_C}[\Phi^C_{\beta,\nu}(X,X')]$, to show that this is also high. By our rounding theorem, Theorem 5.3 we then conclude that the expected value of the Condition&Round algorithm, when performed on C must be high. This corresponds to Step 3(a) in Algorithm 8.1.

(4) Lastly in Lemma 8.10 we show that given a subroutine that finds a subgraph with high Condition&Round value, there is an algorithm that uses this subroutine and finds a high value assignment to the whole graph. This corresponds to the while loop in Algorithm 8.1. Combining this lemma with Lemma 8.9 (discussed above), we get our main theorem.

We prove the theorem below, after establishing each of these components separately in the full version. First let us discuss the structure theorem for Johnson graphs and define the notion of restrictions.

Definition 8.3 (r-restricted subcubes of J)**.** Given an (n,ℓ,α)-Johnson graph J and a set $A \subseteq [n]$ with $|A| = r$ such that $0 \le r \le \ell-1$, we let $J|_A$ denote the vertex-induced subgraph of J induced by vertices that contain the set A. We call such a subset an r-restricted subcube of J. Note that when $A=\emptyset$ and $r=0$, $J|_A$ is defined as the whole graph J.

Definition 8.4 (Restrictions of Functions)**.** For the (n,ℓ,α)-Johnson graph J, given a function $F : V(J) \to \mathbb{R}$ and a set $A \subseteq [n]$ with $A = r$, such that $0 \le r \le \ell-1$, we define the restricted function $F|_A : \binom{[n]\setminus A}{\ell-r} \to \mathbb{R}$ as,

$$F|_A(X) = F(A \cup X).$$

Further, let $\delta_A(F)$ denote the fractional size of the function restricted to the subcube $J|_A$, that is,

$$\delta_A(F) := \delta(F|_A) = \mathop{\mathbb{E}}_{X\sim\binom{[n]\setminus A}{\ell-r}}[F|_A(X)].$$

When $A=\phi$ and $r=0$, we have that $F|_A(X) = F(X)$ for all $X \in \binom{[n]}{\ell}$ and $\delta_A(F) = \delta(F) = \mathbb{E}_\pi[F]$.

We prove that every set in J that is not correlated with any r-restricted cube, has high expansion (as a function of r).

Theorem 8.5 (Structure theorem for Johnson graphs). *For all $\alpha \in \mathbb{Q}$ with $\alpha < \frac{1}{2}$, all integers $\ell \in \mathbb{N}$ and all large enough integers $n \gg \ell$, the following holds: Let J be a (n,ℓ,α)-Johnson graph and π be the uniform distribution over $V(J)$. For every integer r such that $0 \le r \le \ell/2$ and every function F that is not correlated with any r-restricted subcube, F has high expansion (as a function of r):*

$$\{F(X)\in[0,1]\}_{X\in V(J)} \vdash_2 \langle F, LF\rangle_\pi \ge (1-(1-\alpha)^{r+1})\cdot$$

$$\left[\left(1-O_\ell\left(\frac{1}{n}\right)\right)\mathop{\mathbb{E}}_\pi[F] - 8^r\binom{\ell}{r}\left(\sum_{j=0}^{r}\mathop{\mathbb{E}}_{Y\in\binom{[n]}{j}}[\delta_Y(F)^2]\right)+B(F)\right].$$

where $B(F)$ represents the Booleanity constraints and equals $\mathbb{E}_\pi[F^{\circ 2}-F]$.

Let us compare this theorem with [28] and for simplicity let $\varepsilon < 1.9\alpha$. Roughly, the structure theorem in [28] implies that for every non-expanding set S with expansion ε, there exists a 1-restricted subcube C such that the S is large inside C: $|S\cap C|/|C| \ge \Omega(1)$. From this theorem, one can derive the fact that in fact a $\delta(S)/\ell$-fraction of the 1-restricted subcubes have this property, where $\delta(S)$ denotes the fractional size of S (by applying their theorem iteratively). Further this implies that given a distribution D over non-expanding sets, say of the same size δ, there exists a 1-restricted subcube C such that, $\mathbb{E}_D[|S\cap C|/|C|] \ge \Omega(\delta/\ell)$.

But the above line of reasoning is not amenable to a low degree sum-of-squares proof because although each iterative step requires only a constant degree SoS proof, to get the final statement we need to apply the theorem $\Omega(n)$ times and this takes degree $\Omega(n)$. Our final aim is to prove the distribution-version of the statement. Our structure theorem gets around this barrier and directly proves the fact, using a constant degree SoS proof, that given a non-expanding set S with expansion $\le 1.9\alpha$, many subcubes are such that S is large inside them. That is, rearranging Theorem 8.5, as a corollary we have an SoS proof (in the formal indicator variables of membership in S) that $\mathbb{E}_C[|S\cap C|/|C|] \ge \Omega(1/\ell)$. Given this, we can easily derive the implication for distributions by applying an expectation over D to the latter expression and exchanging expectations. Since we give an SoS proof, the statement holds true for pseudodistributions over non-expanding sets S! Lemma 8.9 carries out precisely this kind of an argument, but in more generality.

The proof ideas of Theorem 8.5 are similar to those in [28], hence we defer the proof of this theorem to the full version. We will now show that under this theorem we get an algorithm for UG on the Johnson graph J. We will first formally define the global shift-partition potential on a subgraph.

Definition 8.6 (Global shift-potential restricted to Subgraphs)**.** Let $I=(G,\Pi)$ be an instance of affine unique games over alphabet Σ. For any $\nu,\beta \in (0,1)$ and subgraph H of G, define the *approximate global shift-partition potential restricted to the subgraph H* to be the quantity:

$$\Phi_{\beta,\nu}(X,X')|_H = \sum_{s\in\Sigma}\mathop{\mathbb{E}}_{u\in H}\left[Z_{u,s}\cdot p(\mathrm{val}_u(X))\right]^2,$$

for $Z_{u,s} = \mathbb{1}(X_u - X'_u = s)$,
$\mathrm{val}_u(X) = \mathbb{E}_{(u,v)\in E(G)}[\mathbb{1}(X \text{ satisfies } (u,v))]$, and $p(x)$ the degree-$\tilde{O}(1/\nu)$ polynomial in the family $P_{\beta,\nu}$, described in Theorem 3.2.

Note that the global shift-partition potential measures the size of the global partition inside H, i.e. the $\mathrm{val}_u(X)$ is a function of all the edges in $E(G)$ that are incident on u, not just the edges in H. We will now use the structure theorem for Johnson graphs to get a lower bound on the global shift-partition potential restricted to subcubes C, $\Phi(X,X')|_C$, when the violations of the assignments X and X' are small. The following lemma is analogous to Lemma 6.4

for certifiable small-set expanders and is proved in the same way. The main difference is in the conclusion of the lemma: instead of getting a lower bound on the shift-partition potential of the whole graph, we get a lower bound on the global shift-partition potential restricted to subcubes.

Lemma 8.7. *For all $\alpha \in \mathbb{Q}$ and all $\ell, n \in \mathbb{N}$ with $\alpha\ell \in \mathbb{N}$ and $\ell \ll n$ sufficiently large, the following holds: If I is an affine unique games instance over the (n, ℓ, α)-Johnson graph J, then for all $\beta, \nu \in (0, 1)$ and for every integer $r \in [\ell/2]$, there is an SoS lower bound of the following form on the average of the approximate global shift-partition potential $\Phi_{\beta,\nu}$ over r-restricted subcubes of J:*

$$\mathcal{A}_I \cup \{p \in P_{\beta,\nu}\} \vdash_{\tilde{O}(1/\nu)}$$
$$\sum_{j=0}^{r} \mathop{\mathbb{E}}_{Y \in \binom{[n]}{j}} [\Phi_{\beta,\nu}(X, X')|_{(J|_Y)}]$$
$$\geq \frac{1}{8^r \binom{\ell}{r}} \left(1 - \frac{2\mathrm{viol}(X)}{1 - \beta - \nu} - 2\nu - o_n(1) - K_{\beta,\nu}(X, X')\right),$$

where $\mathcal{A}_I$ are the axioms defined for I by program (1), $\mathrm{viol}(X) = 1 - \mathrm{val}(X)$ is the fraction of constraints X violates, and $K_{\beta,\nu}(X, X') = \frac{1}{1-(1-\alpha)^{r+1}}\left(\mathrm{viol}(X) + \mathrm{viol}(X') + \frac{2\mathrm{viol}(X)}{1-\beta-\nu} + 2\nu\right)$.

Using the lemma above, we can now prove that given a pseudodistribution μ over a highly satisfying instance of unique games over the Johnson graph we can find an r-restricted subcube C with high Condition&Round value. Let us define this precisely:

Definition 8.8 (Condition&Round Value)**.** Given a unique games instance $I = (G, \Pi)$ and a degree 4 shift-symmetric pseudodistribution μ over I, for every subgraph H of G, let $\text{ind-val}_\mu(H)$ denote the expected fraction of satisfied edges when independent rounding is performed on $V(H)$ using the marginals of μ, i.e. $\text{ind-val}_\mu(H) := \mathbb{E}_{(v,w)\sim E(H)}[\sum_s \tilde{\mathbb{E}}_\mu[X_{v,s}]\,\tilde{\mathbb{E}}_\mu[X_{w,\pi_{vw}(s)}]]$. Let the Condition&Round value, denoted by $\text{CR-val}_\mu(H)$ be the value obtained by performing Algorithm 5.1 on H, i.e. $\text{CR-val}_\mu(H) := \mathbb{E}_{u\sim V(H)}[\text{ind-val}_{\mu|X_u=0}(H)]$.

We will show this by first finding a cube C that has high global shift potential, $\tilde{\mathbb{E}}_\mu[\Phi(X, X')|_C]$, using Lemma 8.7 above. We then relate the global shift potential to the shift-partition potential on C, which we will denote by $\Phi^C(X, X')$. The only difference between the two potentials is that the latter is measured using the value of a vertex *inside* C and is the usual definition of the shift-partition potential on the graph C. We show that the subcube C has small expansion, hence we can relate the global value of a vertex (when averaged over all edges in $E(G)$ incident on it) to the local value of a vertex (when averaged over just the edges in $E(C)$ incident on it), thus relating the global shift-partition potential to the shift-partition potential on C. In particular, we will show that there exists C that has high shift-potential; using the analysis of the Condition&Round algorithm, Theorem 5.3, this immediately gives us that there exists an r-restricted subcube that has high Condition&Round value. To find such a cube C algorithmically, one can just enumerate over all r-restricted subcubes in time n^r and check in polynomial time whether C has high Condition&Round value or not. We make this argument formal in the proof of the following lemma:

Lemma 8.9. *For all $\varepsilon \in [0, 0.001)$, for all $\alpha \in \mathbb{Q}$ and $\alpha < \frac{1}{2}$, all integers $\ell \in \mathbb{N}$ with $\alpha\ell \in \mathbb{N}$ and all integers $k, n \gg \ell$ sufficiently large, the following holds: Let I be an affine unique games instance over the (n, ℓ, α)-Johnson graph J with alphabet size $|\Sigma| = k$ and $\mathrm{val}(I) = 1 - \varepsilon$. Then for $r = \lfloor \frac{32\varepsilon}{\alpha} \rfloor$, given a degree-$\tilde{O}\left(\frac{1}{\varepsilon} 2^{4r}\binom{\ell}{r}\right)$ shift-symmetric pseudodistribution μ satisfying the axioms $\mathcal{A}_I$, in time n^r we can find a s-restricted subcube C with $s \leq r$ such that C has high Condition&Round value: $\text{CR-val}_\mu(C) \geq \Omega\left(\frac{\varepsilon}{2^{4r}\binom{\ell}{r}}\right)$.*

Using the above theorem, we can find a subcube C with high value, say $\geq \delta = \Omega_{\ell,\alpha}(1)$, and then perform derandomized Condition&Round algorithm to get a δ-satisfying assignment to the vertices of C. But this may be a negligible fraction of edges of the whole graph (since even a 1-restricted subcube is a $o(1)$-fraction of J), and we need to satisfy $\Omega_{\varepsilon,\alpha,\ell}(1)$ constraints. To achieve this, after setting the vertices of the subcube C, we alter the pseudodistribution μ and apply our algorithm iteratively: we randomize μ on $V(C)$, so that these vertices are completely uncorrelated with any other vertex. This ensures that the value of any edge incident on $V(C)$ is $1/|\Sigma|$, which is much smaller than δ, under the modified pseudodistribution μ'. Then, we run the algorithm again on μ' to find a subcube C' with high Condition&Round value. Since edges that are incident on previously assigned vertices have very low value, we can show that the subcube C' has low intersection with C. Furthermore the subcubes we find have low expansion, so we get that, the derandomized Condition&Round algorithm when performed on $C' \setminus C$ satisfies a constant fraction of edges incident on C'. We continue in this way until the modified pseudodistribution's value drops by $\Omega(\varepsilon)$. We show that at each iteration of the while loop, by modifying the pseudodistribution we lower the value by an amount that is proportional to the fraction of edges we satisfy in that step. Thus, after sufficiently many iterations we lower the value of the pseudodistribution by $\Omega(\varepsilon)$ and hence satisfy an $\Omega_{\ell,\alpha}(\epsilon)$ fraction of the edges in the graph. We make this argument formal via the following lemma:

Lemma 8.10. *Let $\varepsilon_0 \in (0, 1)$ be a universal constant and $\delta : [0, 1] \to [0, 1]$ be a function. Let $\varepsilon < \varepsilon_0/2$ be any constant, and let $\delta_{\min} = \min_{\eta\in[\varepsilon,2\varepsilon]}(\delta(\eta))$. Let G be a regular graph and I be any unique games instance on G with alphabet size $|\Sigma| = k \geq \Omega(\frac{1}{\delta_{\min}})$ and value $1 - \varepsilon$.*

Suppose we have a subroutine $\mathcal{A}$ which given as input μ, a shift-symmetric degree-D pseudodistribution satisfying $\mathcal{A}_I$ with $\mathrm{val}_\mu(I) = 1 - \eta \geq 1 - \varepsilon_0$, returns a vertex-induced subgraph H such that, 1) $\text{CR-val}_\mu(H) \geq \delta = \delta(\eta)$ and 2) the edge-expansion of H is $O(\eta)$.

Then if $\mathcal{A}$ runs in time $T(\mathcal{A})$, there is a $|V(G)|(T(\mathcal{A}) + |V(G)|^{O(D)})$-time algorithm which finds a solution for I that satisfies an $\Omega(\delta_{\min}^2\varepsilon)$-fraction of the edges of G.

This completes the analysis of Algorithm 8.1 modulo the proofs of the lemmas. Combining the lemmas above, Theorem 8.2 easily follows.

ACKNOWLEDGMENTS

Mitali Bafna was supported in part by a Simons Investigator Award and NSF Award CCF 1715187. Boaz Barak was supported by NSF awards CCF 1565264 and CNS 1618026 and a Simons Investigator

Fellowship. Pravesh K. Kothari was supported by NSF CAREER Award 2047933.

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